

# Phase 8J — Operational Dashboard & GIS Visualization

Technical Walkthrough, Emergency Command Center & Leaflet.js GIS Interface

|                                                              |                                                     |
|--------------------------------------------------------------|-----------------------------------------------------|
| Frontend Architecture: SPA (HTML5 / CSS3 / ES6 / Leaflet.js) | Backend Mount: FastAPI StaticFiles at /dashboard/   |
| GIS Vector Layers: 8 NER State Polygons + 73 CWC Radar Pins  | Bilingual Support: English & Tamil (■■■■■■■) toggle |
| Live SIH Scenarios: 6 Preset Demo Locations Toolbar          | Test Suite: 151 / 151 Tests Passed                  |

## 1. Dashboard Features & Layout

- **Map-First Interface:** Centered on Northeast India with CartoDB Dark Matter, OpenStreetMap, and OpenTopoMap basemaps.
- **Dynamic Pin & Radar Pulse:** Click-to-query pin with pulsing concentric radar rings color-coded to risk verdict (Green=LOW, Yellow=WATCH, Orange=HIGH, Red=CRITICAL).
- **50 km Proximity Ring:** Visual circle illustrating CWC telemetry threshold.
- **Operational Decision Panel:** Authoritative risk badge, fusion score bar, static susceptibility, telemetry multi-window accumulations, and explainability reason codes.
- **Bilingual Localization:** Instant toggle between English and Tamil with 100% key parity.

## 2. SIH Presentation Demo Presets

| Preset Scenario                 | Coordinates          | Terrain & Station Context                    | Verdict               |
|---------------------------------|----------------------|----------------------------------------------|-----------------------|
| Guwahati Urban (Assam)          | 26.1445°N, 91.7362°E | Gentle slope (1.2°), CWC station 4.2 km away | LOW RISK              |
| Tawang Alpine (Arunachal)       | 27.5925°N, 91.6087°E | Steep slope (31.4°), CWC > 50km (NO_DATA)    | WATCH (Precautionary) |
| Mangan / Chungthang (Sikkim)    | 27.5028°N, 88.5284°E | High relief (>50m std), fragile Himalayas    | HIGH RISK             |
| Cherrapunji / Sohra (Meghalaya) | 25.2986°N, 91.7317°E | Extreme rainfall escarpment plateau          | WATCH (Rainfall)      |
| Aizawl Ridge City (Mizoram)     | 23.7271°N, 92.7176°E | Inhabited ridge crest, shale sequences       | HIGH RISK             |
| New Delhi (Domain Rejection)    | 28.6139°N, 77.2090°E | Outside 8 NER states bounding box            | HTTP 400 REJECT       |